Juan David Muñoz-Bolaños
Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

ASML
Hiring Team
De Run 6501
5504 PE Veldhoven, Netherlands

Subject: Application for Optical / System Engineer position

Dear Hiring Team,

I am writing to express my strong enthusiasm for an Optical/System Engineer position at ASML. As a Photonics Engineer concluding my PhD in Innsbruck, I want to leverage my technical expertise in adaptive optics, high-performance microscopic optical systems, and computational pipelines to contribute to ASML's cutting-edge lithography equipment. I am excited about the opportunity to push the boundaries of semiconductor manufacturing through precision engineering and advanced optics.

My research is deeply rooted in correcting complex optical aberrations and managing light-matter interactions. Designing and implementing closed-loop optical systems has provided me with hands-on experience in managing high-power lasers, wavefront control, and precise optical alignment, which directly translates to the multidisciplinary challenges of ASML's optical modules and EUV/DUV systems.

Adaptive Optics & Wavefront Control:
ASML's lithography systems require exceptional control over metrology and projection optics. Over the past few years, I have architected adaptive optics setups using spatial light modulators and precision optomechanics to mitigate severe aberrations. My expertise in diagnosing and dynamically correcting wavefront errors maps directly to the precision optical tuning required for next-generation systems.

High-Precision metrology & Low-Light Detection:
Achieving nanometer-level precision demands robust metrology and sensor technology. In my work with advanced nonlinear microscopy (two-photon and Raman), I have optimized signal-to-noise ratios in photon-starved regimes. This experience has honed my ability to design sensitive detection pathways, analogous to the highly sensitive wafer alignment and inspection sensors developed at ASML.

System Integration & Computational Control Pipelines:
I bridge the gap between optical hardware and software. Through my PhD and previous industry experience as a Machine Vision Engineer, I have built hardware control algorithms and GPU-accelerated pipelines in Python for high-performance data processing. My capability to integrate industrial cameras, illumination strategies, and intelligent computational systems makes me well-suited for improving and optimizing complex, multi-component optical machines.

About me, I am a Colombian living in Innsbruck with a German Master's in Photonics, and an upcoming PhD in optical microscopy and adaptive optics. After finishing my PhD, I will get an Austrian graduate working permit that makes the hiring process seamless to you, equivalent to an EU citizen. I am available after April/May and I am open to chat about how my expertise in optics and solving problems can support ASML's mission. Thank you for your time and consideration.

Sincerely,

Juan David Muñoz-Bolaños
Optical System Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer | Adaptive Optics

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145
juan.munoz@i-med.ac.at | LinkedIn Profile

PROFESSIONAL SUMMARY

Optical Systems Engineer specializing in Adaptive Optics (AO) and high-precision optical setups. Expertise in diagnosing and correcting complex optical aberrations, directly transferable to precision lithography and metrology systems. Background spans nonlinear optics, hardware-software integration, GPU-accelerated pipelines, and machine vision.

TECHNICAL COMPETENCIES

Optical Engineering & Adaptive Optics
Wavefront control: Design of closed-loop AO systems using spatial light modulators, lasers, and precision optomechanics. Experience in optical modeling, system alignment, aberration correction, and mitigating optical distortions.

Precision Detection & Imaging
Nonlinear optics & Spectroscopy: Managing signal-to-noise ratios in photon-starved regimes. Hardware integration: Photomultipliers (PMTs), spectrometers, single-mode fibers, and advanced imaging detectors.

Control, Data Analysis & Machine Vision
Computing: High-performance image processing pipelines using Python (JAX/CuPy) for GPU acceleration. Systems Integration: Industrial cameras, illumination control, and intelligent machine vision. Embedded: FPGA (LabView) hands-on for real-time control.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Engineering
Jan 2022 – Present

- **Adaptive Optics:** Engineered closed-loop control systems for optical microscopes to correct severe optical aberrations dynamically.
- **Advanced Optical System Design:** Implemented Two-Photon microscopes, balancing high-power laser delivery with highly efficient, sensitive detection pathways.
- **Lab Management & Precision Alignment:** Maintained high-precision optical infrastructure, ensuring strict operational standards and continuous system optimization.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Assistant Researcher
Jun 2020 – Dec 2021

- **System Prototyping:** Assembled and integrated vibrational imaging prototypes utilizing spectrometers, fiber probes, and CW lasers.
- **Data Analysis:** Processed large datasets using machine learning methods to identify weak signal patterns from high-noise environments.

INTECOL S.A.S. — Medellin, Colombia
Machine Vision System Engineer
Mar 2018 – Mar 2019

- **Optical Metrology & Tracking:** Developed industrial quality control systems utilizing computer vision, industrial cameras, and custom illumination.
- **Systems Integration:** Managed product lifecycles and physical deployment for safety-critical automated inspection systems.

EDUCATION

Medical University of Innsbruck — Austria
Ph.D. in Adaptive Optics & Microscopy — *Jan 2022 – Present*

Friedrich Schiller University Jena — Germany
M.Sc. in Photonics — *2019 – 2021*

University of Cauca — Colombia
B.Sc. in Physics Engineering — *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards:** ZEISS Summer School (2025), COLFUTURO Grant (2019).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. Applied Sciences, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte — Medical University of Innsbruck
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft — Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Prof. Dr. Christian Spielmann *Dekan*	Prof. Dr. Silvana Botti *Vorsitzende des Prüfungsausschusses*

REPUBLICA DE COLOMBIA

PASAPORTE
PASSPORT

Tipo / *Type*	Cod. país / *Country code*	Pasaporte Nº / *Passport No.*
P	COL	AV410884

Apellidos / *Surname*
MUÑOZ BOLAÑOS

Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*
04 JUN/JUN 1994

Núm. personal / *Personal No.*
CC1061772278

Sexo / *Sex*
M

Lugar de nacimiento / *Place of birth*
LA CRUZ COL

Fecha de expedición / *Date of issue*
11 DIC/DEC 2018

Autoridad / *Authority*
G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*
10 DIC/DEC 2028

Firma del titular / *Holder's signature*

ORD AV410884 10 DIC/DEC 2028

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SONSTIGE ANGABEN
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AUFENTHALTSTITEL
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NACHNAMEN Vornamen
MUÑOZ BOLAÑOS
Juan David
GESCHLECHT
STAATSANGEHÖRIGKEIT
GEBURTSDATUM
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04 06 1994
KARTE GÜLTIG BIS
16 05 2026
ART DES AUFENTHALTSTITELS
NIEDERLASSUNGSBEWILLIGUNG
SONSTIGE ANGABEN
FORSCHER
349461
RESIDENCE PERMIT

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

ÖSD ist Mitglied von

ALTE

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Dr. Thomas Marschner
Program Manager Funding Projects

Dr. Thomas Stammler
CTO & Head of Product Strategy

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020

F. Sondermann

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest

Leibniz ipht
LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

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Jena, 13 November 2020

Prof. J. Popp
Chairman
of Leibniz-IPHT

F. Sondermann
Executive Member
of Leibniz-IPHT

Juan David Munoz Bolanos
Guest